TUSHAR MADAN WATTI

9322920594 | watitushar@gmail.com | Linkedin | Github

EDUCATION

B.Tech, Computer Engineering (Software Engineering**)** | Vishwakarma Institute of Technology, Pune | 2028 | CGPA: 8.43
Diploma, Computer Engineering | Government Polytechnic, Nagpur | 2025 | 77.20%
SSC | Adarsh Sanskar Vidyalaya | 2022 | 88.60%

EXPERIENCE

Full-Stack Developer Intern | RB Tech Services | Ahmednagar, Maharashtra | Jan 2025 – Apr 2025

- Developed and deployed an E-Learning platform supporting 10+ courses, 50+ video lectures, 100+ study notes, and quizzes.
- Implemented backend modules for user authentication and course enrollment using PHP and MySQL.
- Designed and integrated a relational database with 5+ tables to support course and user data management.
- Built responsive web functionality using HTML, CSS, JavaScript, PHP, and jQuery/AJAX.

PROJECTS

G4B (Glasses for Blind's) | Python, YOLOv7, OpenCV, Roboflow, ESP32-CAM, Arduino, Computer Vision

- Developed an AI-based assistive system using YOLOv7, ESP32-CAM, sensors, speakers, and computer vision to support mobility for visually impaired users.
- Annotated 3,217 images in Roboflow and trained a custom 6-class object detection model.
- Achieved 99.4% mAP@50, 99.5% precision, 99.9% recall, and 99.7% F1 score on the object detection task.
- Integrated real-time object detection, distance estimation, and audio feedback with an ultrasonic sensor backup system.

Smart AI Load Balancer | Python, FastAPI, Scikit-learn, Machine Learning

- Engineered an AI-based load balancing system using Python and FastAPI to dynamically route requests across distributed server instances.
- Trained a Random Forest model using simulated CPU load, latency, and active connection telemetry, achieving 99.0% predictive routing accuracy.
- Implemented AI-based routing alongside traditional load-balancing strategies for comparative performance evaluation.
- Built a real-time dashboard to visualize server load, request routing, latency, and system performance.

Restaurant IQ | Python, Scikit-learn, Pandas, NumPy, Machine Learning

- Developed an AI-powered risk analysis system to classify restaurant business viability using a dataset containing 200,000+ restaurant records.
- Applied data preprocessing, exploratory analysis, and machine learning techniques to financial, operational, and customer metrics.
- Built predictive models to classify restaurant risk categories and support analysis of future market trends.
- Developed an interactive web application for real-time risk predictions, model comparison, and decision support.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, PHP, HTML, CSS
Frameworks & Libraries: FastAPI, Flask, React.js, Node.js, jQuery/AJAX, Pandas, NumPy, Scikit-learn, OpenCV
Databases: MySQL
AI/ML & Computer Vision: Machine Learning, Random Forest, YOLOv7, Computer Vision, Roboflow
Hardware & Embedded: ESP32-CAM, Arduino
Core Concepts: Object-Oriented Programming, REST APIs, Backend Development, Database Design, Load Balancing

LEADERSHIP & EXTRACURRICULARS

- Directed a school Scout Parade by coordinating team activities and execution; earned the Best Parade award for team discipline.